

# Aanidhya Patidar

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## SUMMARY

Computer Science (AI & ML) student focused on scalable backend systems and applied machine learning. Recently built and deployed an agricultural fintech PWA with custom AI localization, real-time WebSocket chat, and sub-zooms API pipelines serving Indian farmers. Seeking a backend or ML internship to solve complex engineering problems.

## EDUCATION

### Vellore Institute of Technology

Bachelor of Technology in Computer Science (AI & ML)

GPA: 8.71/10.0

Bhopal, MP

2023 – 2027

## TECHNICAL SKILLS

**Languages:** Python, C++, JavaScript (ES6+), SQL

**Frameworks & Libraries:** FastAPI, Flask, React, Vite, Pandas, NumPy, Scikit-learn, LightGBM, PyTorch, SQLAlchemy, TanStack Query, i18next, Recharts, Sentence-Transformers

**Databases:** PostgreSQL, SQLite, MySQL

**Developer Tools:** Git, GitHub, VS Code, Postman, Alembic, Docker, Vercel, Render

## PROJECTS

### Krishi Khata — Digital Farm Ledger & Market Intelligence PWA | *FastAPI, React, Vite, PostgreSQL, Gemini AI*

2025 – 2026

- Built and deployed a mobile-first PWA for Indian farmers — digital accounting ledgers, real-time mandi price tracking, and AI crop advisory. Frontend on Vercel, backend on Render, backed by Neon PostgreSQL.
- Designed a JIT mandi price pipeline using government APIs, caching 30-day commodity histories per district in PostgreSQL for sub-zooms lookups — replacing wasteful batch scraping entirely.
- Integrated a bilingual AI engine via Google Gemini — the backend intercepts `Accept-Language` headers and rewrites Gemini system instructions at runtime, streaming native Hindi or English responses without storing translated text.
- Set up a hardened WebSocket gateway for real-time community chat with JWT-authenticated handshakes, pre-upgrade rejection of unauthorized connections, and SlowAPI rate limiting on write-heavy routes.
- Localized the React frontend across 70+ UI strings via i18next, added searchable Combobox dropdowns for mandi queries, and built a Quick-Select Crop Hub that auto-fills search parameters from the farmer's active crops.

### DeepGuard — Deepfake Face Detection System | *Python, PyTorch, EfficientNet-B4, OpenCV, MTCNN*

2025 – 2026

- Built a binary deepfake classifier by fine-tuning EfficientNet-B4 on FaceForensics++ — a benchmark dataset of real and AI-manipulated facial images covering face-swap and reenactment forgery methods.
- Implemented an MTCNN-based face extraction pipeline in OpenCV to crop and align face regions from raw images before classification, isolating the model's focus to facial features and reducing background noise.
- Applied Grad-CAM to generate class activation heatmaps, surfacing which facial regions (mouth corners, eye edges, blending boundaries) the model weighted most heavily — making predictions interpretable beyond a confidence score.
- Handled class imbalance via weighted cross-entropy loss and used stratified splits across forgery types to prevent the model from overfitting to a single manipulation method.
- Evaluated on held-out FaceForensics++ test set using AUC-ROC alongside accuracy to account for threshold sensitivity, with Grad-CAM overlays surfacing consistent artifact patterns at compression boundaries.

## CERTIFICATIONS

- Applied Machine Learning in Python** — University of Michigan (Coursera)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**

## LANGUAGES

English (Fluent), Hindi (Native)